

ALPACA AI TRADING AGENTS HACKATHON 2026

AURIGA

Autonomous Quant Research & Investment Agent

Discover, validate, deploy -- the LLM narrates, it never decides.

Submission deck | 5-year backtest | Alpaca paper \$100k

github.com/keisary/AURIGA

THE PITCH

Rules over vibes. The quant system decides. The LLM explains.

MARKET DATA

5y · 1H/1D · options

ML ENGINES

A1 + A2 + A3 · XGBoost

RISK ENGINE

8 gates · fail-closed

ALPACA MLEG

paper · atomic multi-leg

LLM --> narrates / interprets (no execution authority)

9

rules admitted

~0.7

5

vol shock AUC

8

risk gates

25

large-cap universe

Three agents, three testable theses

A1 DIRECTION

Directional alpha

XGBoost -> explainable rules

Debit vertical spreads (bull / bear)

9

rules admitted

Sharpe ≥ 2 , WR ≥ 0.65 , PF ≥ 1.5 , ≥ 30 trades, BH/FDR-controlled

A2 VOLATILITY RISK

Volatility risk signal

XGBoost classifier on vol shock

No execution -- a guard for the risk engine

~0.75

AUC on $RV[t+24] > 1.5 \times RV[t]$

Long-vol straddles measured at median Sharpe -2.6. Buying vol loses -- so A2 stops premium sales when it flags danger.

A3 PREMIUM SELLER

Gated income engine

AVOID_SELL rules on the ~2% tail

Put / call credit spreads (~3% OTM, 5% wide)

~98%

of periods structurally profitable

Stops selling when overbought + drawdown + regime shift coincide -- exactly the rare tail ML can flag.

8 deterministic gates -- fail-closed

Every order passes ALL gates. If a gate cannot be verified, the order is blocked.

G1

Daily loss limit

-2% of equity --> no new orders

G2

Max exposure / asset

10% of capital

G3

Max exposure / sector

25% of capital

G4

Max total exposure

80% of capital

G5

Max positions

12 concurrent positions

G6

Vol danger (A2)

$P(\text{vol shock}) > 0.35$ --> block premium sales

G7

AVOID_SELL (A3)

any triggered rule --> block premium sales

G8

Liquidation

total P&L < -25% --> close positions

LLM NEVER DECIDES -- it only narrates facts after the deterministic engine has acted.

03 / ALPACA INFRASTRUCTURE

Built on Alpaca paper. Real market data. Atomic option spreads.

MARKET DATA

IEX feed + option chains

- 5 years of 1H / 1D bars
- Live option chains (OCC symbols)
- Universe: 25 US large caps + ETFs
- 36 features (technical + quantitative)

EXECUTION

MLEG multi-leg orders

- alpaca-py + multi-leg Level 3
- Vertical spreads as atomic orders
- Zero commissions on paper
- Bracket TP/SL + buying-power checks

CADENCE

Research -> signal -> risk

- Research cycle (A1 + A2 + A3)
- Current-signal check + spread picker
- Risk gates -> paper orders
- Daily LLM narrative + Streamlit

ING -> FEAT -> DISC -> VAL -> SEL -> OPT -> RSK -> EXEC -> NAR

Four things that are hard to copy

01

Explainable alpha

Every position traces to a human-readable rule with backtested metrics. Not a black box -- you can read why AURIGA is in a trade.

02

Honest research

We measured what does NOT work (long volatility: median Sharpe -2.6) and re-purposed A2 as a risk signal. The system adapts to evidence, not vibes.

03

Two orthogonal income streams

Directional debit spreads (trend, A1) + gated premium selling (range / theta, A3) -- one hedges the other in most regimes.

04

Institutional-grade risk

8 deterministic gates + ML risk flags before any order. Fail-closed by design: any gate that cannot be verified blocks the order.

Stated honestly -- not marketing

L1

Options backtest is a synthetic proxy

Alpaca free tier has no historical option prices. Option P&L in backtests is estimated via Black-Scholes repricing ($IV = RV + \text{risk premium}$). Live paper fills (real option quotes) are the ground truth.

L2

A final 20% temporal holdout is kept untouched

During strategy discovery and admission, we keep a virgin 20% tail of the data. Metrics shown are from train + validation; holdout results are not claimed yet.

L3

5 days of P&L is a small sample

Results during the competition period are indicative, not statistically conclusive. Paper trading research -- not investment advice.

Paper trading research only -- not investment advice.

"AURIGA doesn't ask an LLM whether to trade."

The quantitative system decides. The LLM explains.

EXPLAINABLE

Every rule is human-readable

DETERMINISTIC

Risk gates before any order

EVIDENCE-DRIVEN

Measures what does not work

Repo --> github.com/keisary/AURIGA

Thanks -- looking forward to the jury's questions.